



Assessment Cover Sheet

Unit Code and Title	ICT203 Web Application Development
Assessment Name and Number	Assessment 3 Full-Stack Web Application with Responsible AI Integration
Teaching Period / Year	S.Y 2026
Lecturer / Unit Coordinator	Dr Israel Fianyi
Submission Date	29 - 05 - 2026
Submission Type	Group

Team Members (Complete ONLY for group assessments)

Student Name	Student ID	Contribution (%)	Responsibilities for the Assessment
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Janceen Piasidad	250087	100%	Team Lead

Declaration

By submitting this assessment, I/we confirm that this work is my/our own and complies with the CIHE Academic Integrity Policy and the QA70 Use of GenAI in Teaching, Learning, Assessment and Research Policy. Any use of AI or other tools has followed the assessment instructions and guidance provided in the CIHE Student Guide on the Use of AI and has been fully acknowledged in the AI/Tool Use Declaration Table below. I/we understand that inappropriate or unacknowledged use of AI, or submission of work that is not my/our own, may constitute academic misconduct. I/we also confirm that this assessment has not been previously submitted unless approved, and I/we consent to its use for standard academic integrity, quality assurance, and AI governance review processes.

Student(s) Signature:

Date: 29 - 05 - 2027

GenAI Tool	Description
Chatgpt	For formatting reports, organizing the various sections, checking grammar, improving wording, and structuring the presentation.
Gemini AI	Used to help with formatting suggestions, proofreading, and the help of clarifying written content.

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I. Project Proposal

I. A. Problem Statement

Through university clubs and associations, student involvement, student communication, student leadership, and participation are promoted. There are lots of student organizations that still use paper-based systems and social media in the way of communicating with each other about membership and events. It indicated that digital platforms facilitate enhancing student engagement and interaction in educational settings (Bond et al., 2021). Bergdahl et al. (2020) also said that the online engagement systems are used to facilitate communication and participation in digital learning communities, and as stated by Ayanwale (2025), digital platforms enhance accessibility, collaboration, and user engagement. Hence, it is necessary to create the University Club and Association Website that will be central to enhancing communication, organization, accessibility, and students' involvement in the university.

I. B. Target Users

The construction of the University Club and Association website will help students and the overseers of a club to run and coordinate their group activities, communications, and attendance at events. Henne et al. (2022) mentioned that digital platforms provide accessibility, communication, and participation in education by having a centralized online system. The principal users of the system are: **Students** who wish to be involved in clubs and events, and be updated on the activities going on at the university. **Club Officers**, who handle memberships, announcements, and club activities;

Event Coordinators, who keep track of the schedules, registration, and participation; **Volunteers and Participants**, persons with whom the university is involved in any program or activity and who should receive announcements and information about various events. The system will offer a common platform for enhancing communication, organizing and managing students' engagement in the university.

I. C. Features of the System

The objective of the University Club and Association Website is to enhance the management of membership, announcements, and participation in events at the different university clubs. The system offers digital solutions that help to communicate, organize, provide access, and involve students within the university community. System features are: Online club membership registration and tracking (Students); Provide a secure login procedure for the administrator/members; Regular announcements and information on the central platform; Ability to create events, manage and register participants; Simple event and member record-keeping and storage; Increased communication with students, club officers, and event coordinator(s); easy/user-friendly interface for university users.

I. D. Technology Stack

The development of the proposed University Club and Association Website is expected to follow a full stack web development principle for the benefits of efficiency, scalability and maintainability.

Using **HTML (HyperText Markup Language)**, **CSS (Cascading Style Sheet)** and **JavaScript**, the front end will provide a responsive and interactive user interface for students and club officers. The backend will be built using **PHP** or **Node.js** with Express, responsible for user authentication, system functions, and communicating with the database. Data like user accounts, memberships, events and announcements will be safely stored and easily managed in **MySQL**. The coding, version control and team collaboration will be facilitated with the tools found in **Visual Studio Code** and **GitHub**.

I. E. Team roles

Name:	Role:	Responsibilities:
Carl Anthony Cababat	Front End Developer	Responsible for the interface and layouts of the web application to be built, and a responsive design that includes navigation and validation of forms.
Rene Jane Arciga	Backend Developer and Database Designer	Developing server-side functionality, authentication, APIs and handling communication between the front end and database. Knows how to design and maintain a

		database structure such as tables, relationships, and integrity.
Sumana Nagarkot	QA tester/ Documentation Member	Knows how to design and maintain a database structure such as tables, relationships, and integrity.
Janceen Piasidad	Team Lead	Oversee and manage the entire project, including task assignment, tracking schedule, and meeting deadlines of project deliverables.

I. F. Milestones

During **Week 4 - 5**, there will be a period of planning and design, which will include the collection of system requirements, project proposal, system architecture design, and database ERD design. In **Weeks 6 - 7**, the first part of the system development will be to create the basic functionality of the website such as simple role based access control, database connection, user logins etc. The core features of the event and announcement management, search capability and user interface improvements will be a focus for **week 8-9**. In **Week 10**, AI capabilities like content or event description generation will be added to the system and will be subject to human review prior to output being saved. During **Weeks 11 - 12**, the team will perform functional,

usability, and security testing to discover and solve issues that the system may have. Lastly, the project will be documented, and the system will be ready for implementation.

II. System Design

II. A. Architecture diagram (Client-server-DB)

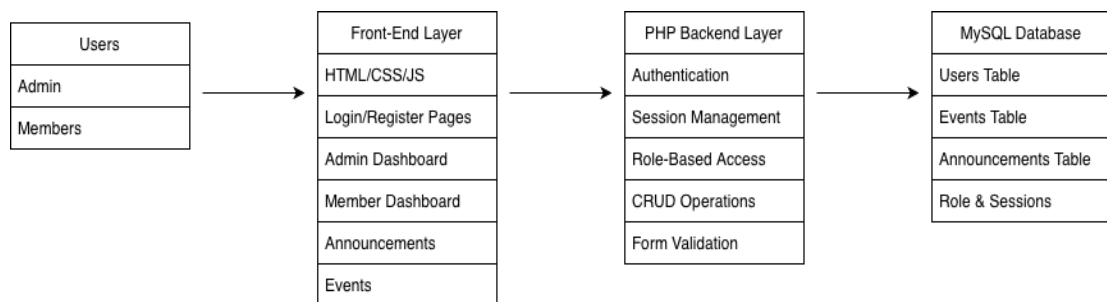


Figure 2.1: Architecture Diagram

The general architecture of the University Club and Association Management System is shown in figure. There are two major users of Admin and Members who log in the system and interact with the system using the front-end part which is created by using HTML, CSS and JavaScript. Front-end layer offers vital functions corresponding to login and registration web pages, dashboard, announcements, and management of events. The PHP backend layer then processes the user requests, such as authentication session management, role-based access control, CRUD operations and form validation to provide secure, efficient and functioning of the system. lastly all the system data including user information, events, announcements, roles and session data are stored and managed within the MySQL database.

II. B. ERD (Database schema)

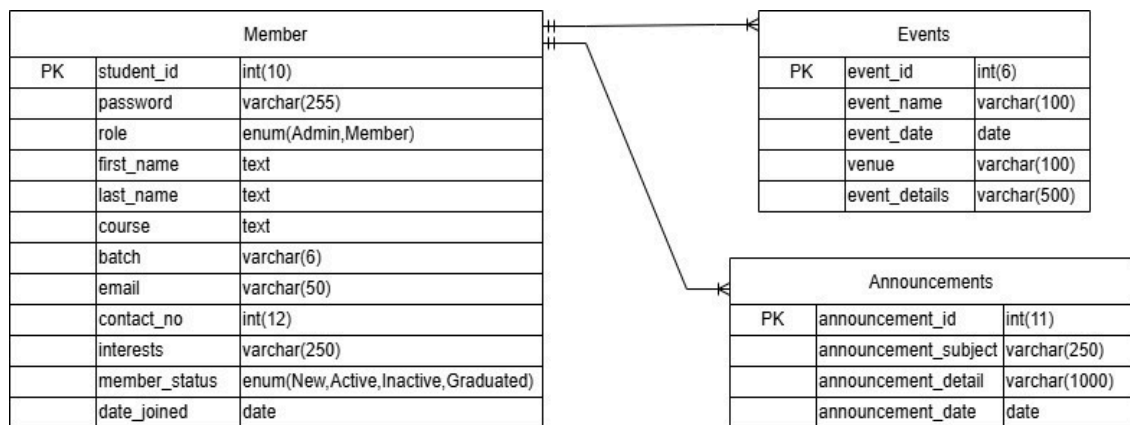


Figure 2.2 Entity Relationship Diagram (ERD)

Figure illustrates the Entity Relationship Diagram (ERD) of the proposed Database Schema for University Club and Association Website. The database consists of three main entities: Member, Events, and Announcements, which are connected through one-to-many relationships. User information is kept in the Member entity, and the Events and Announcements entities track event information and what information is important to relay within the system. This ERD will facilitate an efficient organization and maintenance of the data for three major purposes: keeping track of members, communicating with them, and managing events.

III. Security & Risk Register

With the introduction of the University Club and Association Management System, there are several security and operational concerns that need to be addressed to ensure the reliable operation and protection of the system. The table below summarises the identified risks and the measures to mitigate.

Risk	Description	Impact	Mitigation Strategy
SQL Injection Vulnerabilities	Attackers can try to use inputs (e.g., login forms or search bars) to alter the queries that are sent to the database.	High - May lead to unauthorized access, modification, or deletion of database records.	Ensure the use of prepared statements (parameterised queries) and do not join queries together just by concatenating the strings.
Unauthorised Access to Restricted Areas	Users would be trying to gain access to pages that are reserved to the admin or users would be tryin to do actions for which they have no authority.	High - Inability or inability to monitor use adequately and/or chances to the system or exposure of sensitive information.	Use role-based access control, (RBAC) and secure session validation on both the front and back end components.
Insecure Password Handling	Weak storage attribute (Users may be compromised due to weak passwords or	High - May lead to entering the user's account and misuse of	Use full password options and secure hashing passwords, e.g. bcrypt or

	poor password storage methods.	personal information.	password_hash.
Cross-Site Scripting (XSS) Attacks	User input (such as announcement or event description) can include malicious scripts.	Medium - may disrupt user sessions or cause to appear manipulated content to users.	Validate user inputs and sanitise and escape everything submitted by a user.
Inaccurate AI-Generated Content	If not carefully monitored, AI-generated content can include incorrect or misleading information.	Medium - Could reduce the reliability and accuracy of system content.	Uses a human-in-the-loop editing process; have users manually edit and review the content created by AI before saving.

IV. Test Evidence

IV. A. Test Cases 10 (mix of positive and negative)

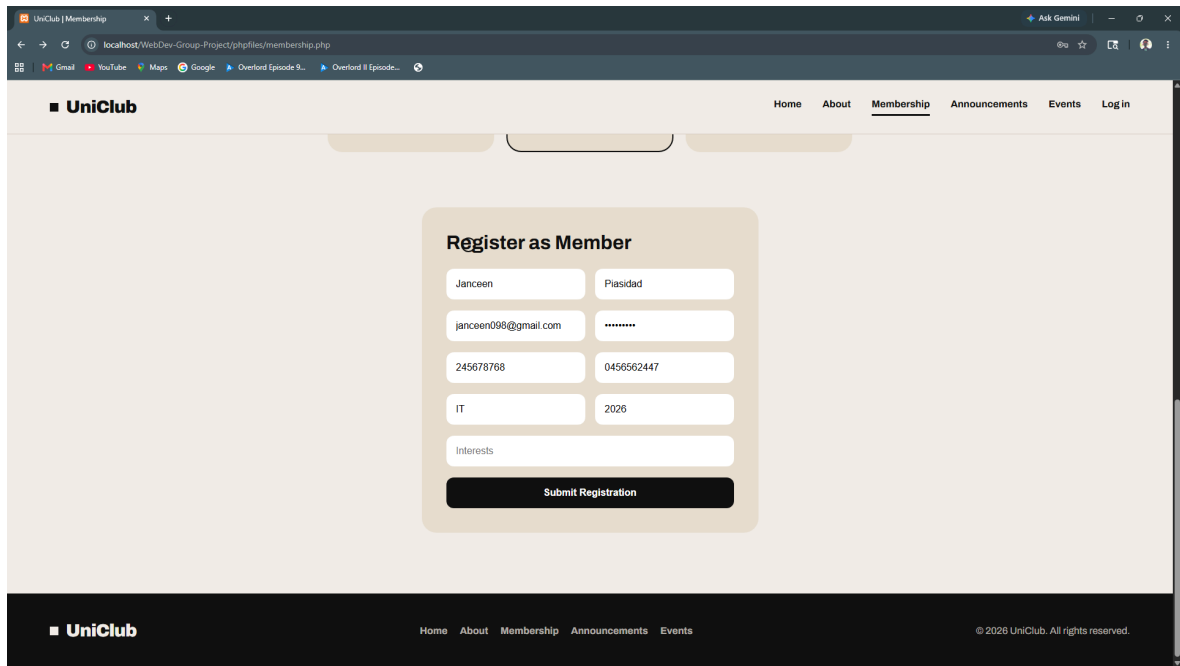
Test #	Positive Test Cases	Negative Test Cases
1	User Registration: Enter a	User Registration: Leave

	valid username, email, and password, then click Register	required fields empty and submit form.
2	Login System: Enter correct username and password	Login System: Enter incorrect password
3	Create Announcement: Admin fills announcement form correctly and submits	Role-Based Access: Standard user tries to access admin page directly via URL. Access denied or redirected to login page
4	Create Event: Admin fills an Event form correctly and submits.	SQL Injection Protection: Enter ' OR 1=1 - into login field
5	Delete Announcement: Admin clicks delete on an announcement	Duplicate Member ID: User register with an already registered ID

IV. B. Evidence Screenshots

Positive Evidence:

Test 1 - Valid Registration Details

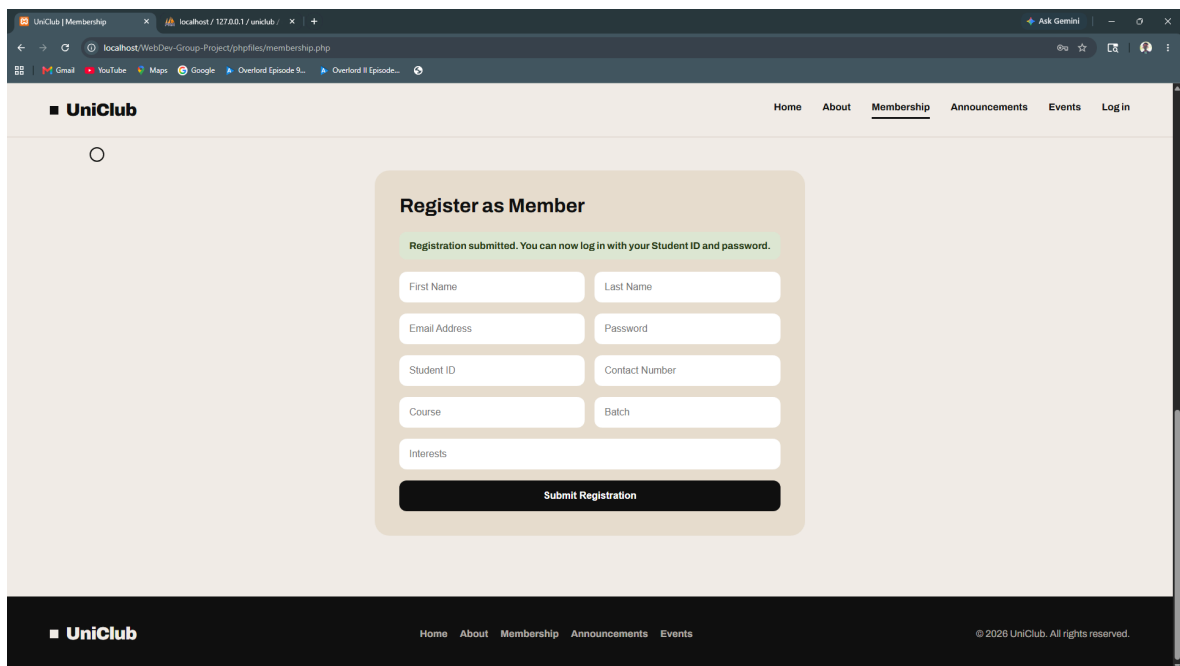


The screenshot shows a web browser window with the URL `localhost/WebDev-Group-Project/phpfiles/membership.php`. The page has a navigation bar with links: Home, About, Membership, Announcements, Events, and Log in. The main content area features a "Register as Member" form. The form fields are filled with the following data:

Field	Value
First Name	Janceen
Last Name	Piasidad
Email Address	janceen098@gmail.com
Password	*****
Student ID	245678768
Contact Number	0456562447
Course	IT
Batch	2026
Interests	

Below the form is a "Submit Registration" button. The footer of the page includes the UniClub logo, navigation links, and a copyright notice: "© 2026 UniClub. All rights reserved."

Valid Registration Message



The screenshot shows the same UniClub Membership page, but now a success message is displayed: "Registration submitted. You can now log in with your Student ID and password." The "Register as Member" form is still visible, but the fields are empty. The "Submit Registration" button remains at the bottom of the form. The navigation bar and footer are identical to the previous screenshot.

Database Validation

Test 2 - Valid Login Credentials

UniClub | Log In

Home About Membership Announcements Events Log in

Log In

2024984028

Log In

UniClub

Home About Membership Announcements Events

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Admin Logged In without any issues. This is the same with a non-admin user.

UniClub | Admin Profile

Home About Announcements Events Manage Announcements Manage Events Profile Logout

Club dashboard, Andrea.

Review your profile details and keep an eye on recent membership activity from one place.

Andrea Sachs
Journalism
Admin

Admin Details

STUDENT ID 2024984028	EMAIL andy.sachs@yahoo.com
CONTACT NO 498365920	BATCH S22028
DATE JOINED 2025-05-05	

UniClub

Home About Membership Announcements Events

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Test 3 - Create Announcement

UniClub Home About Announcements Events Manage Announcements Manage Events Profile Logout

Membership Registration Reminder	Students are encouraged to complete membership registration before the end of the month.	2026-05-24
Its my Birthday	Its my birthday Aye	2026-05-26
Test Announcement	Testing	2026-05-26
Club Meeting This Friday	Members are invited to attend the monthly planning and activity meeting.	2026-05-28

Create an announcement

Subject

Announcement

Create Announcement

UniClub Home About Membership Announcements Events © 2026 UniClub. All rights reserved.

Admin Announcement view updated.

UniClub Home About Announcements Events Manage Announcements Manage Events Profile Logout

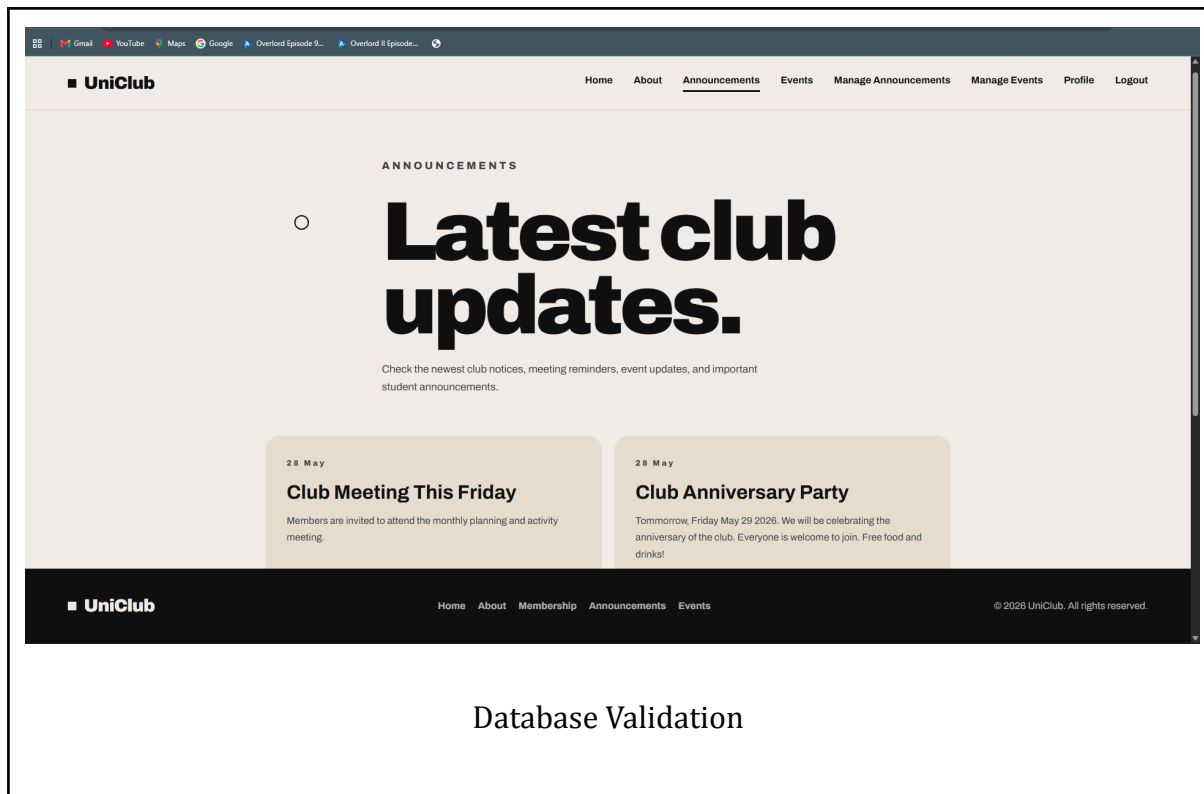
Subject	Announcement	Date
Careers Fair Registration Open	Students can now register for the Annual Careers Fair happening this semester.	2026-05-20
Membership Registration Reminder	Students are encouraged to complete membership registration before the end of the month.	2026-05-24
Its my Birthday	Its my birthday Aye	2026-05-26
Test Announcement	Testing	2026-05-26
Club Meeting This Friday	Members are invited to attend the monthly planning and activity meeting.	2026-05-28
Club Anniversary Party	Tommmorrow, Friday May 29 2026. We will be celebrating the anniversary of the club. Everyone is welcome to join. Free food and drinks!	2026-05-28

Create an announcement

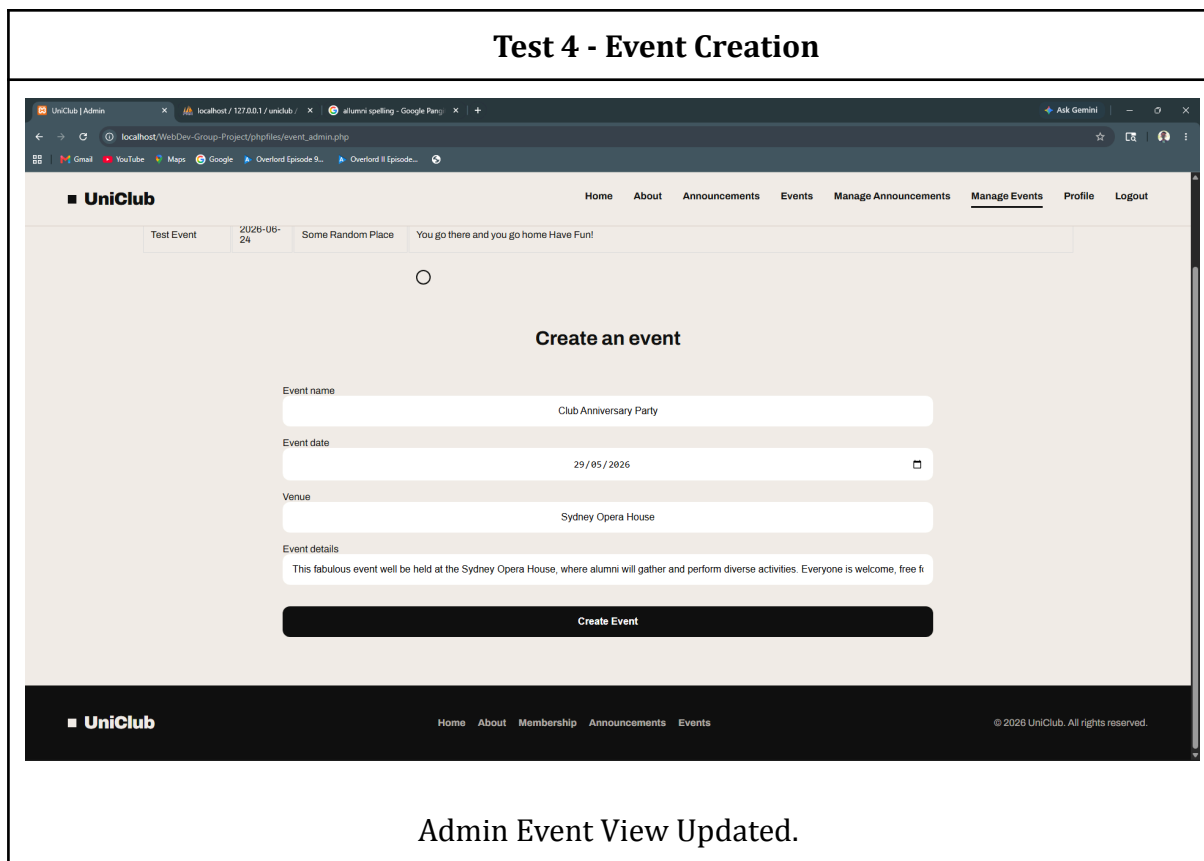
Subject

UniClub Home About Membership Announcements Events © 2026 UniClub. All rights reserved.

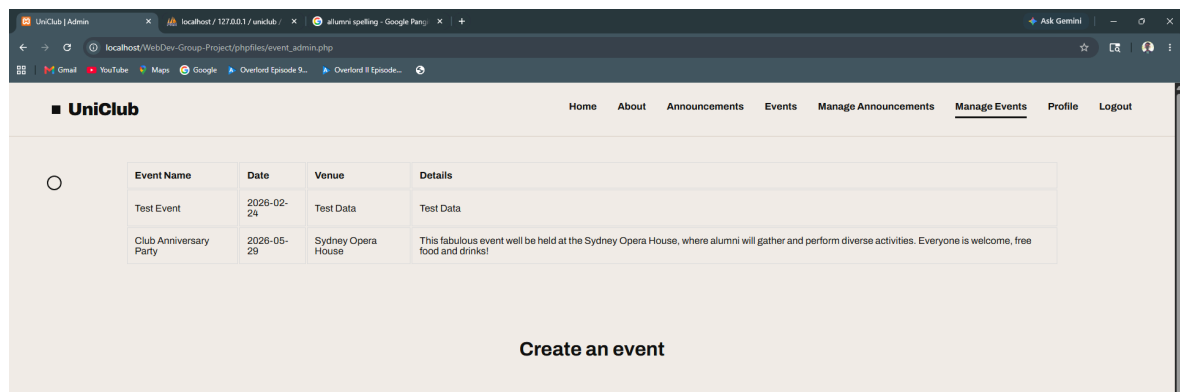
All user announcement pages, Updated.



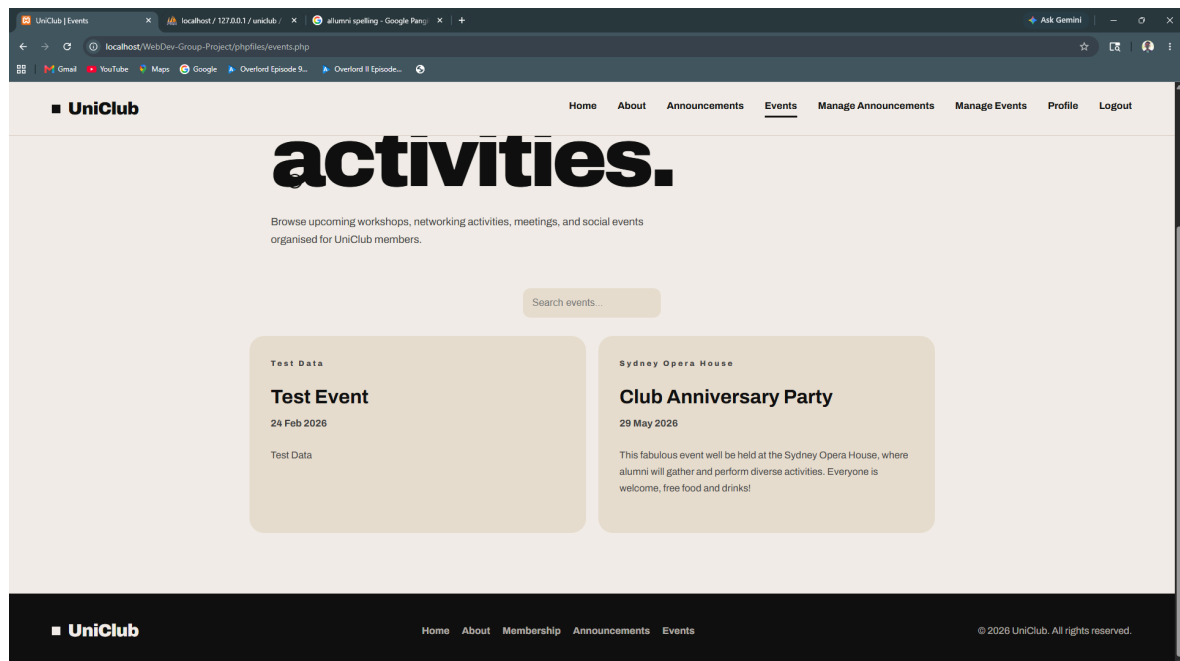
Database Validation



Admin Event View Updated.

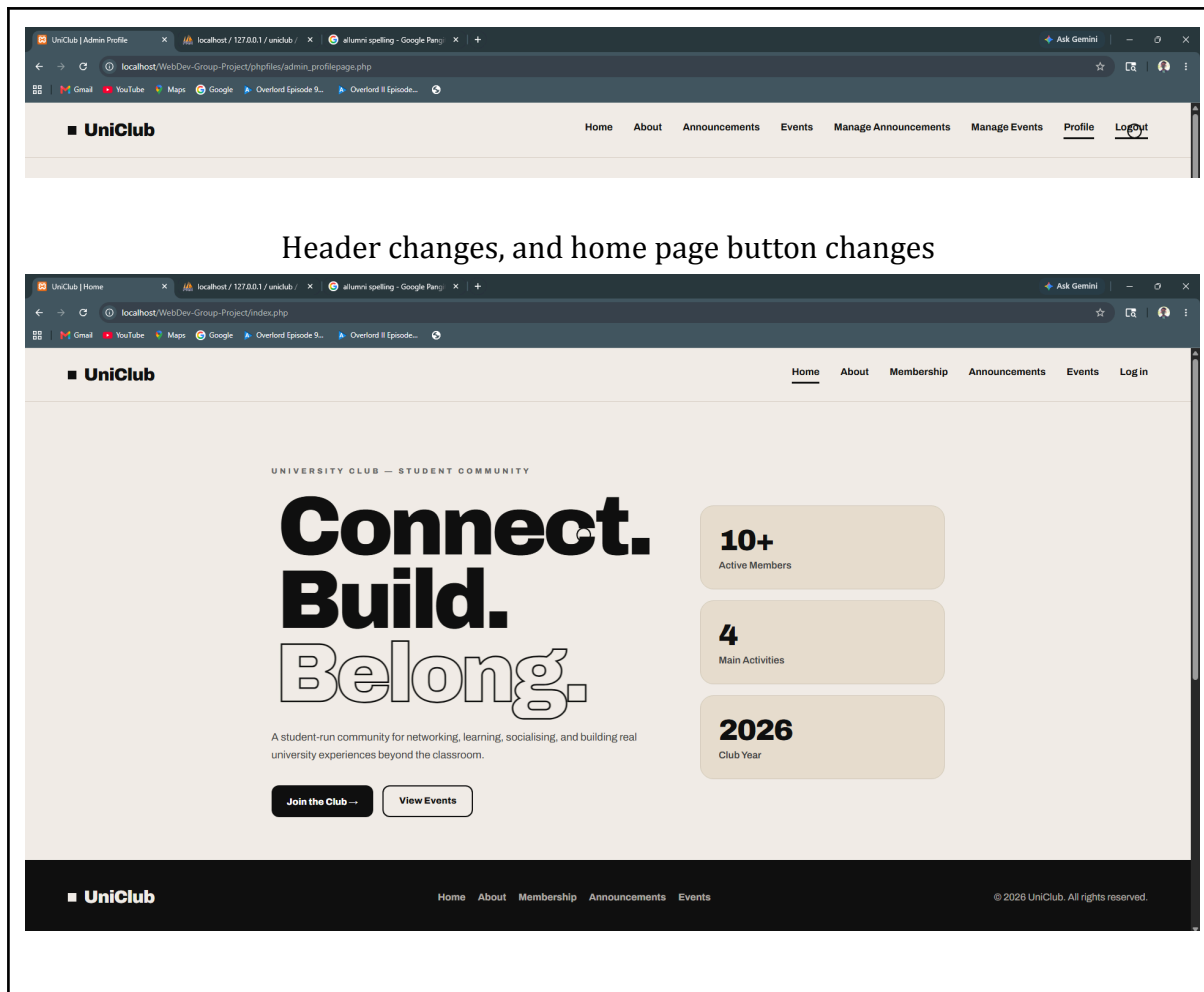


All user Event page is updated.



Database Validation

Test 5 - Session End on logout



Negative Evidence:

Test 1 - User Register while leaving empty fields.

Register as Member

First Name

Please fill out this field.

Last Name

Password

Student ID

Contact Number

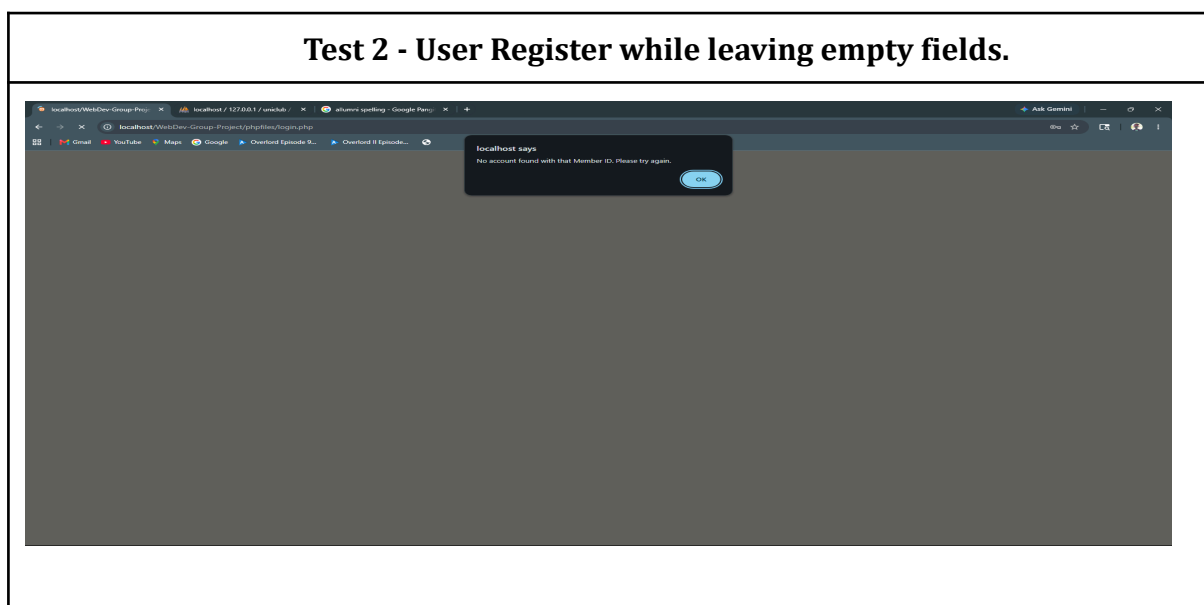
Course

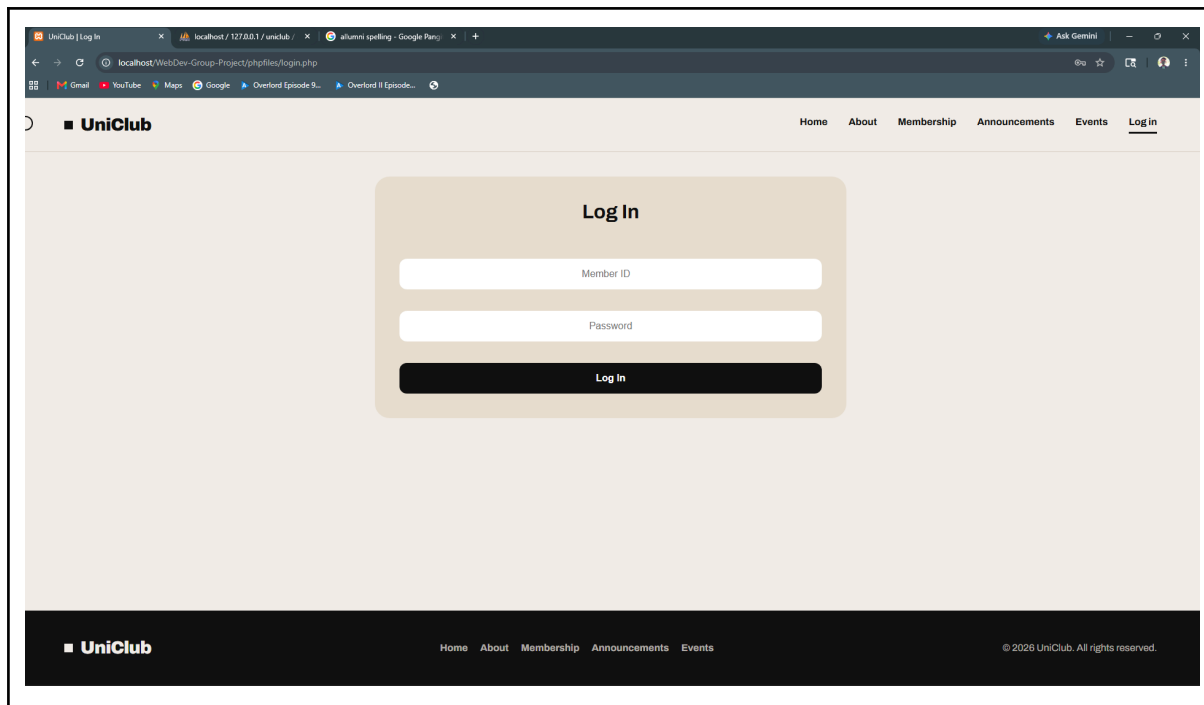
Batch

Interests

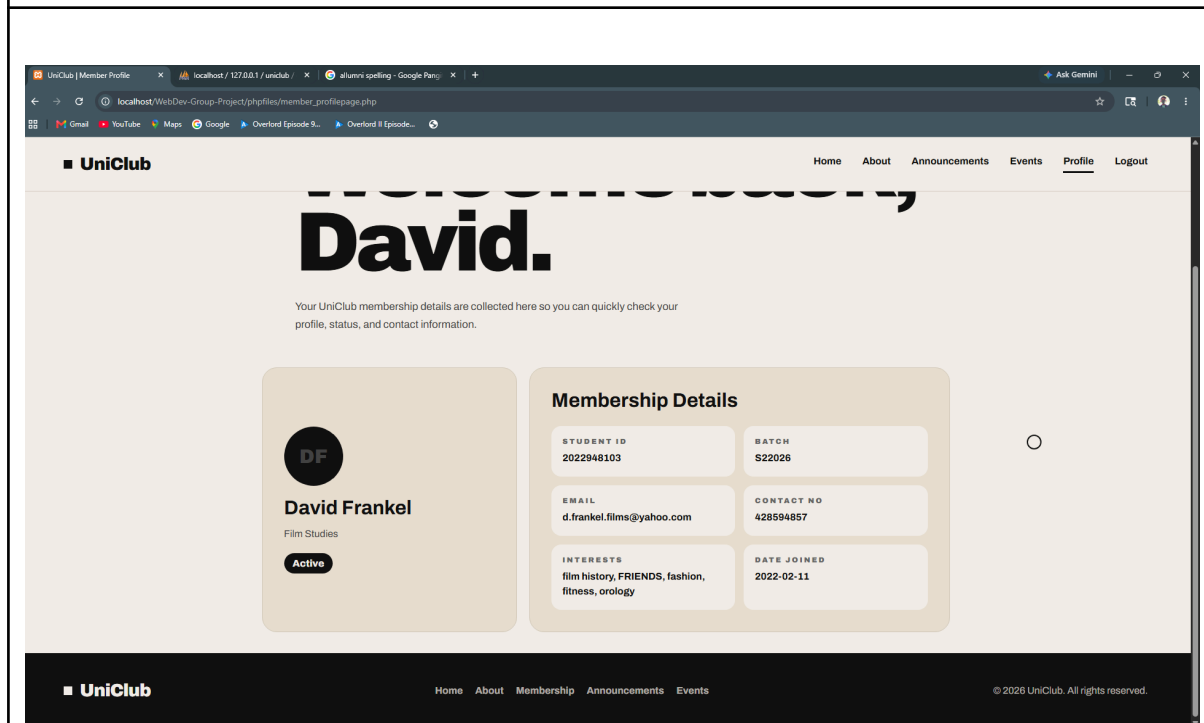
Submit Registration

This will apply to all fields except for the interest field.

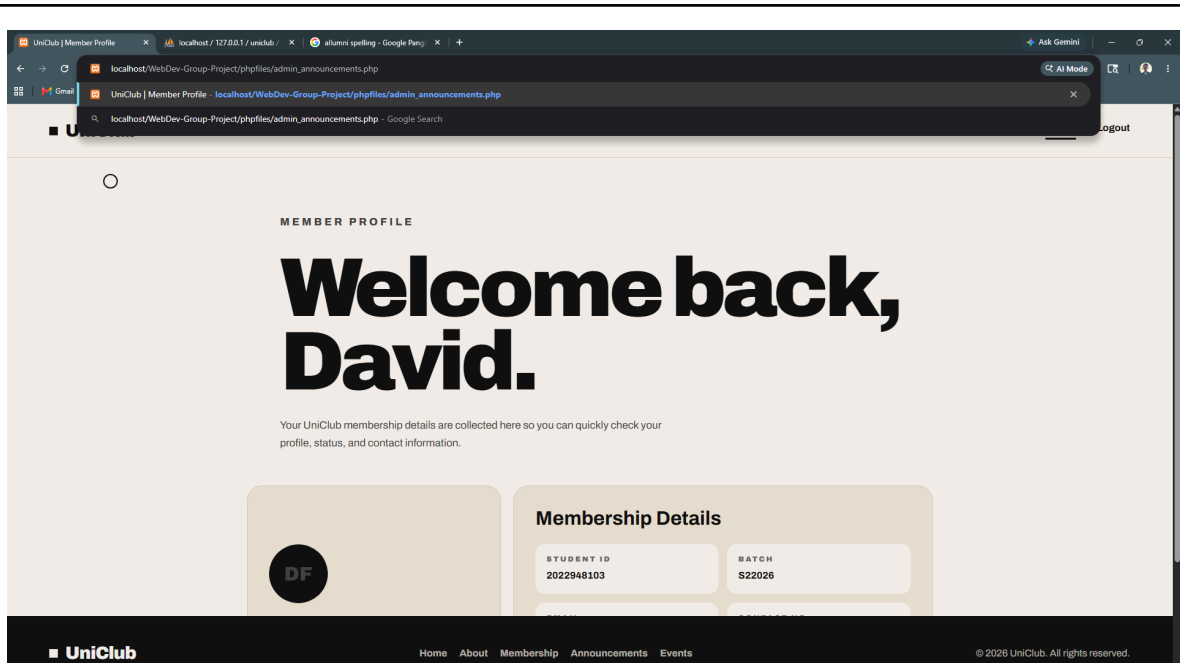




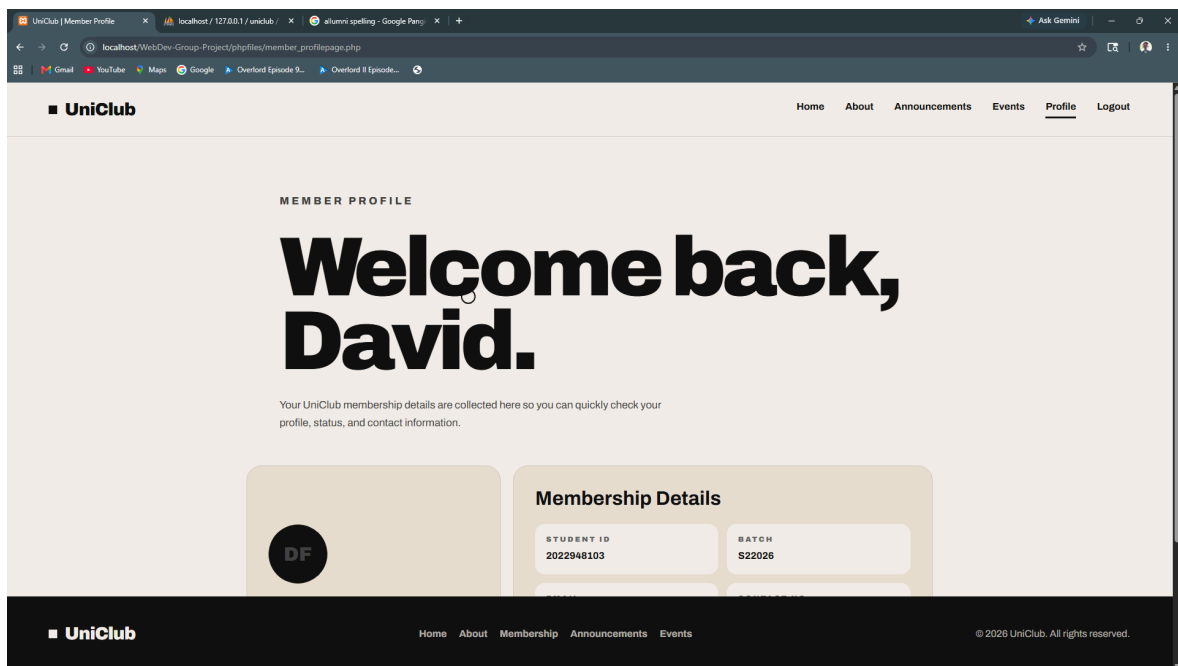
Test 3 - Non Admin member tries accessing admin pages using links



Logged in as member, no admin links in Header.



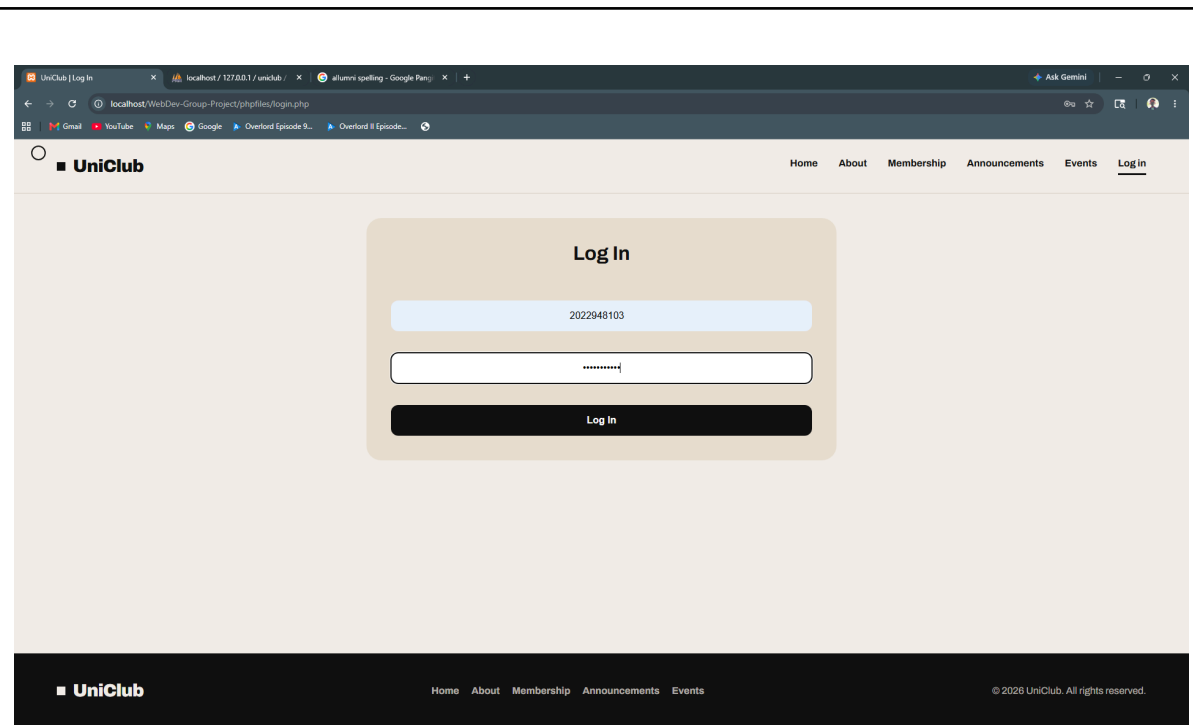
Access admin pages through URL



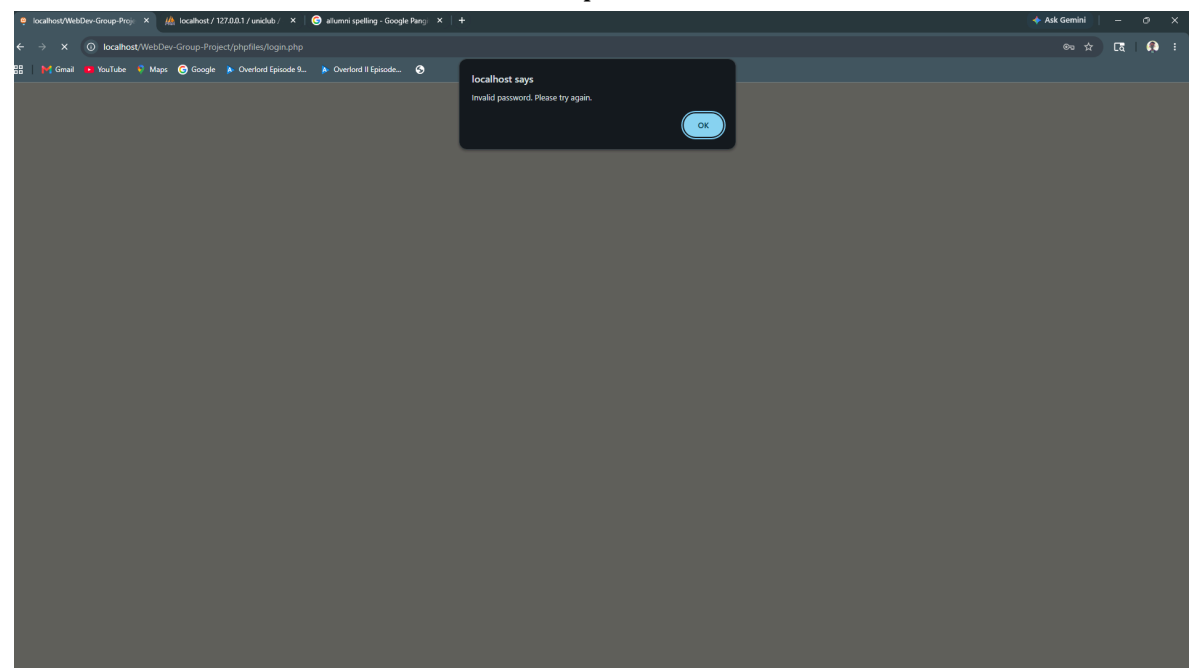
Routed back to Profile Page

Test 4 - SQL Injection

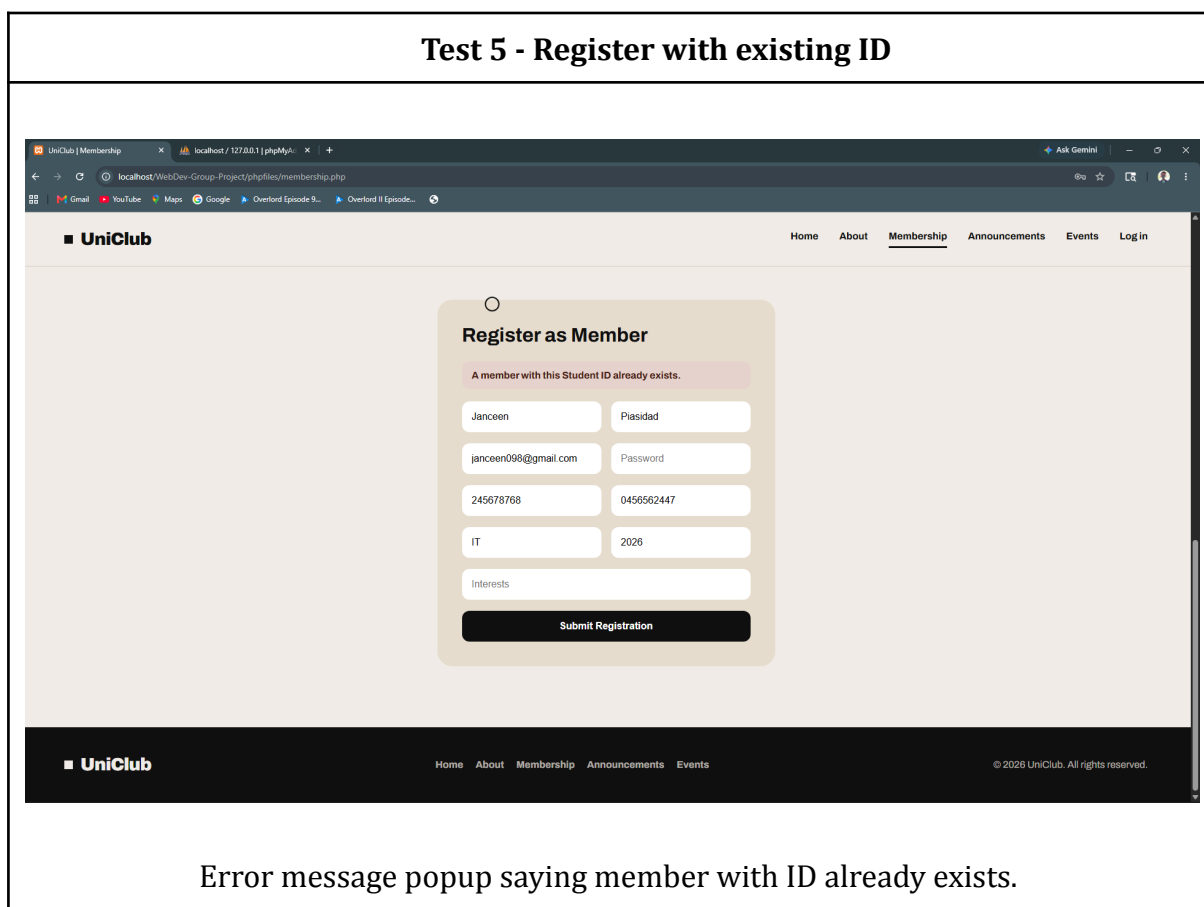
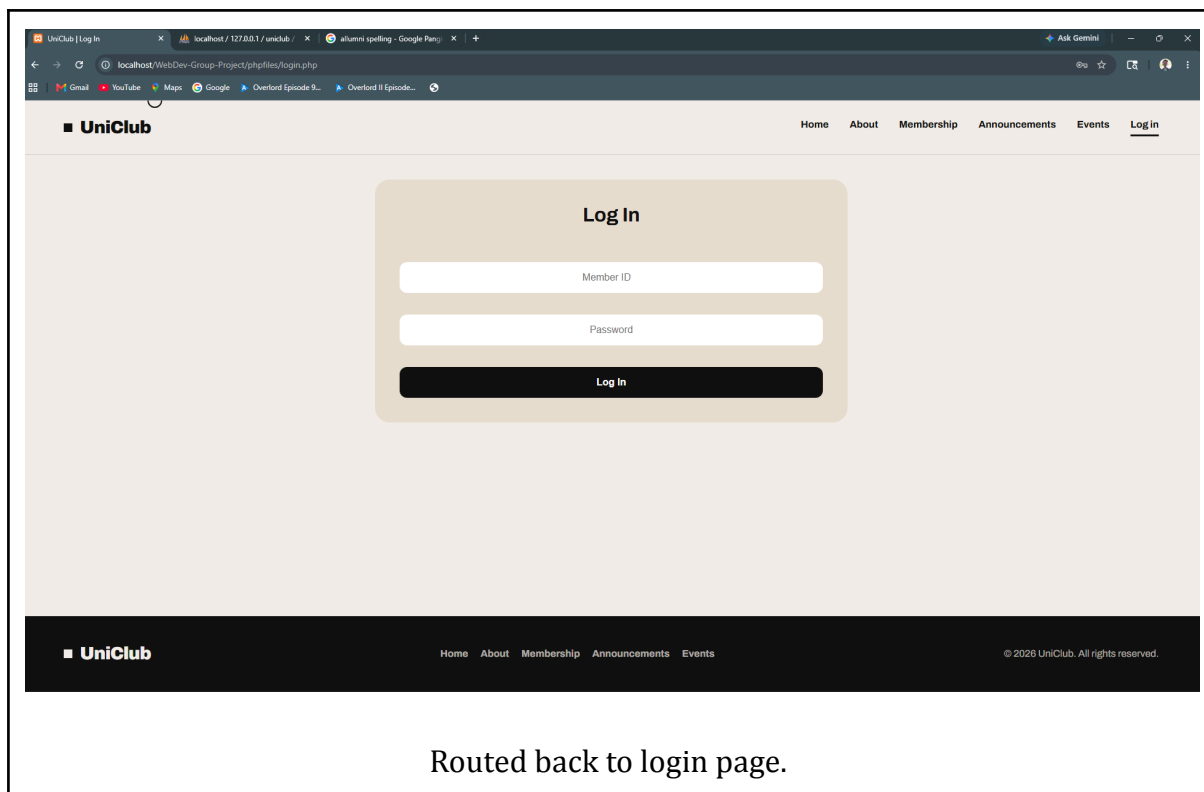
CIHE Australia
116 Pacific Highway North Sydney NSW 2060 Australia
Phone (61 2) 9955 0488 Email: info@cihe.edu.au
ABN 22 611 573 301 TEQSA Provider ID: PRV 14301 CRICOS Provider Code: 03744B



Valid Username and password is 'OR 1=1 –.



Invalid alert



V. AI Governance Appendix

V. A. AI Feature Chosen

It contains a straightforward AI-assisted capability to aid learners in composing and improving content about university club activities. This is essentially used for event descriptions, event announcements, or titles. This is done to save time when creating content while giving the user full control because the AI is not for decision-making; it is only for drafting purposes.

As part of developing the AI, it offered suggestions on how to use better code practices, such as the file layout.php for header/footer templating that could be reused, and the file db.php for database connectivity code that could be reused. These tips contributed to making the system more consistent, avoiding repetition, and maintaining general easy access and manageability.

V. B. Data Handling

The AI capability is limited to simple text that users input, like event information or drafts. No need to give or process any personal or sensitive information. To ensure that data usage is safe:

- To text input into the AI is just what's needed.
- There are no passwords, personal information or private information involved
- Note that output from the AI is not automatically stored in the system.
- The users are required to check content before storing them.

V. C. Prompt Approach

Simple structured prompts are used to guide the AI. Users enter a simple description that includes the name, purpose, and date of the event, and the description get reformatted into a description that is more clear and polished.

The prompts are intended to:

- Give answers that are appropriate to the answer being provided
- Be professional and remain uniformed.
- Avoid any output that is not absolutely necessary or is too complicated

This is important when creating content that doesn't need too much editing

V. D Human-in-the-Loop Mechanism

There is always a human in the loop for end-to-end decision-making. AI-generated content is used as a draft, not a final product.

It works in the following way:

1. The AI generates a draft based on user input
2. User reads the results, able to edit results
3. User has the option to alter, approve or deny it
4. Only the final approved version is saved.

This means that nothing goes without human checking.

V. E. AI Limitations and Mitigation

Limitation	Description	Mitigation
Inaccurate output	Can occasionally produce text that is inaccurate or out of context.	Required user review before saving.
Limited context understanding	AI may not be able to grasp certain details of the club's requirements.	Use clear and structured prompts.
Inconsistent tone	Responses don't necessarily follow a standard structure	Human editing before publication
Overuse of AI	Users rely on AI	Keep AI as an optional tool.
Privacy risks	Give access to vulnerable information.	Restrict use the AI with data that are sensitive to privacy violations.

The system also has a basic AI-powered tool that assists users in developing and refining text for university club events, announcements, titles, and more. This feature is not meant for users to rely on AI and let it create content for them, but rather to make their job of creating content faster while maintaining full control over it. It also recommended improved coding style during development, such as using layout.php for reusable headers and footers, and db.php for reusable database connection.

These suggestions led to increased consistency, decreased duplication of code and an easier to maintain and manage system.

Reference:

- Ayanwale, M. A., Molefi, R. R., Kurata, L., Agunbiade, A. I., Olatunbosun, S. O., & Sanni, T. (2025a). From platform to pedagogy: reimagining student engagement in the digital learning landscape. *Cogent Education*, 12(1).
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<https://doi.org/10.1080/2331186x.2025.2580756>
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